Capstone Project I

Mutual Fund Analytics Platform

Final Project Report

Submitted By:
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Project Start Date:
20 June 2026

Technology Stack:
Python | Pandas | NumPy | SQLite | SQLAlchemy | Jupyter Notebook | Power BI | Financial Analytics

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1. Project Overview

The Mutual Fund Analytics Platform is an end-to-end data analytics project designed to analyze mutual fund performance, investor behaviour, market trends, and portfolio risk factors.

The project focuses on building a complete analytics workflow starting from raw financial datasets to meaningful business insights through automated data pipelines, SQL database management, statistical analysis, advanced risk modelling, and interactive dashboard visualization.

The main objective of this project is to help investors and financial analysts compare mutual fund schemes, evaluate risk-adjusted returns, understand investment behaviour, and identify suitable funds based on investment preferences.

The system includes data engineering, exploratory data analysis, performance analytics, risk measurement techniques, and business intelligence dashboards.

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2. Project Objectives

The key objectives of the Mutual Fund Analytics project are:

• Build an automated ETL pipeline for mutual fund datasets.
• Clean and preprocess raw financial data.
• Design a structured SQLite analytical database.

• Perform exploratory data analysis on NAV, AUM, SIP, and investor trends.
• Calculate financial performance metrics.
• Measure investment risk using advanced statistical methods.
• Build a fund recommendation system based on risk profile.
• Develop an interactive Power BI dashboard for decision-making.

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3. Dataset Description

The project uses multiple mutual fund datasets containing information related to schemes, NAV history, investors, portfolio allocation, and market benchmarks.

Dataset	**Description**
fund_master.csv	Contains mutual fund scheme information including fund house, category, plan, benchmark, and risk details
nav_history.csv	Historical NAV values of mutual fund schemes
investor_transactions.csv	Investor transaction records including SIP, Lumpsum, and Redemption details
scheme_performance.csv	Fund return, expense ratio, and performance indicators
aum_by_fund_house.csv	Asset Under Management information by AMC
monthly_sip_inflows.csv	Monthly SIP investment trends
category_inflows.csv	Mutual fund category wise inflow data
portfolio_holdings.csv	Sector and asset allocation information
benchmark_indices.csv	Market benchmark data including Nifty indices

The datasets were used together to create a complete mutual fund analysis ecosystem.

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4. ETL Pipeline and Data Cleaning

An automated ETL (Extract, Transform, Load) pipeline was implemented using Python.

The pipeline performs extraction of raw CSV files, applies cleaning rules, validates financial values, and loads processed data into SQLite database.

Data Cleaning Operations Performed:

NAV History Cleaning

• Converted date columns into datetime format.
• Sorted records using AMFI code and date.
• Forward filled missing NAV values caused by weekends and holidays.
• Removed duplicate records.
• Validated that NAV values are greater than zero.

Investor Transaction Cleaning

• Standardized transaction categories:

- SIP
- Lumpsum
- Redemption

• Fixed transaction date formats.
• Removed invalid transactions.
• Verified amount values greater than zero.
• Validated investor KYC status values.

Scheme Performance Cleaning

• Converted return columns into numeric format.
• Checked missing and abnormal return values.
• Validated expense ratio range between 0.1% and 2.5%.

Other Dataset Cleaning

Additional datasets including AUM, SIP inflows, category inflows, folio count, benchmark data, and portfolio holdings were cleaned by:

• Removing duplicates
• Formatting dates
• Standardizing text columns
• Validating numeric values

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5. Database Design

SQLite database was designed using a star schema approach to support analytical queries.

Database Name:

bluestock_mf.db

Tables Created:

Dimension Tables:

dim_fund

Stores mutual fund information including:

- AMFI Code
- Scheme Name
- Fund House
- Category
- Risk Type

dim_date

Stores date related information for time-series analysis.

Fact Tables:

fact_nav

Stores daily NAV information.

fact_transactions

Stores investor transaction details.

fact_performance

Stores return and risk metrics.

fact_aum

Stores Asset Under Management details.

Primary key and foreign key relationships were created using AMFI code and date fields.

SQL analytical queries were developed for:

• Top funds by AUM
• Monthly NAV trends
• SIP growth analysis
• State-wise transactions
• Expense ratio comparison

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6. Exploratory Data Analysis (EDA)

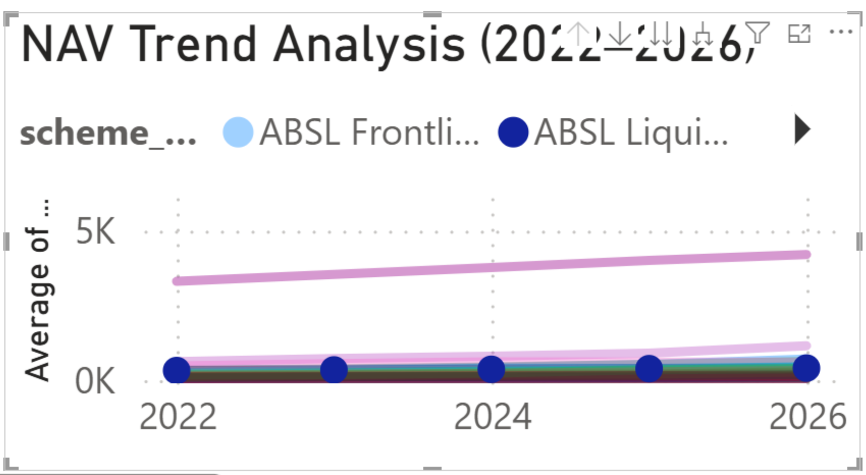
Exploratory Data Analysis was performed using Python libraries including Pandas, Matplotlib, Seaborn, and Plotly.

Analysis Performed:

NAV Trend Analysis

Daily NAV movement of all mutual fund schemes was analyzed from 2022 to 2026.

The analysis helped identify fund growth patterns and market impact periods.

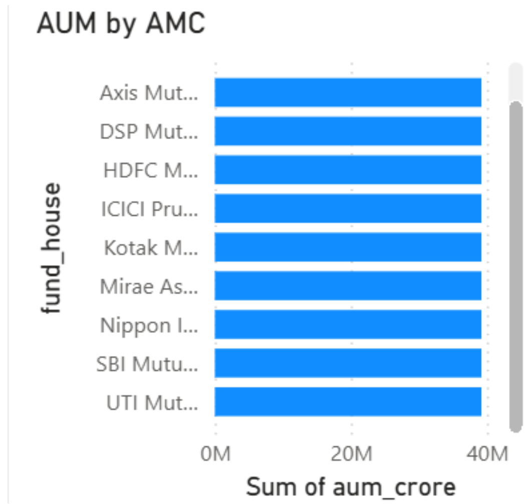


AUM Growth Analysis

Fund house wise Asset Under Management growth was analyzed.

Key observation:

Large AMCs maintained higher asset concentration compared to smaller fund houses.

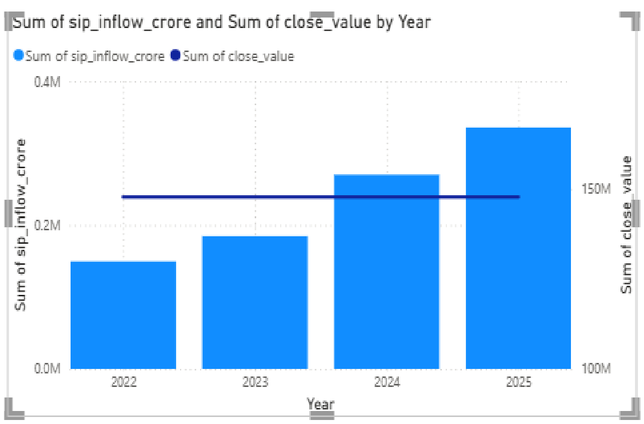


SIP Trend Analysis

Monthly SIP inflow patterns were studied.

Finding:

SIP investments showed consistent growth indicating increasing retail investor participation.



Category Inflow Analysis

A heatmap visualization was created to compare category wise investment flows.

Finding:

Equity-oriented categories showed stronger inflow trends.

category	2024	2025	**Total**
ELSS	4627	1453	**6080**
Flexi Cap	47551	16438	**63989**
Gilt	7753	2642	**10395**
Hybrid	29711	9157	**38868**
Large & Mid Cap	43169	14583	**57752**
Large Cap	19449	6184	**25633**
Liquid	346328	104947	**451275**
Mid Cap	41116	14196	**55312**
Sectoral/Thematic	78107	25722	**103829**
Short Duration	41859	13671	**55530**
Small Cap	34204	12392	**46596**
Value/Contra	12744	4236	**16980**
Total	**706618**	**225621**	**932239**

Investor Behaviour Analysis

Investor demographics were analyzed based on:

- Age groups
- Geographic distribution
- Investment amount
- Transaction preference

The analysis provided insights into investor participation patterns.

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7. Performance Analytics

Advanced financial performance metrics were calculated for mutual fund schemes.

Metrics Implemented:

CAGR

Compound Annual Growth Rate was calculated for:

- 1 Year
- 3 Years
- 5 Years

It helped measure long-term investment growth.

Sharpe Ratio

Sharpe Ratio was calculated using:

Risk Free Rate = 6.5%

It measured risk-adjusted returns of mutual fund schemes.

Higher Sharpe Ratio indicates better return generation for the risk taken.

Sortino Ratio

Sortino Ratio was calculated by considering only downside volatility.

This helped identify funds with lower negative risk.

Alpha and Beta

OLS regression was performed against Nifty 100 benchmark.

Alpha measured excess return generation.

Beta measured market sensitivity of funds.

Output generated:

alpha_beta.csv

Maximum Drawdown

Maximum Drawdown analysis identified the largest decline from peak NAV value.

This helped measure downside risk.

Fund Scorecard

A composite fund ranking model was created.

Score Components:

• 30% CAGR Rank
• 25% Sharpe Ratio Rank
• 20% Alpha Rank
• 15% Expense Ratio Rank
• 10% Drawdown Rank

Output:

fund_scorecard.csv

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8. Advanced Analytics and Risk Metrics

Advanced risk analytics were implemented for deeper investment insights.

Historical Value at Risk (VaR)

Historical VaR at 95% confidence level was calculated.

VaR represents possible downside loss under normal market conditions.

Output:

var_cvar_report.csv

Conditional Value at Risk (CVaR)

CVaR measured the expected loss beyond the VaR threshold.

It provided better understanding of extreme downside scenarios.

Rolling Sharpe Ratio

A 90-day rolling Sharpe Ratio was calculated.

This showed changes in risk-adjusted performance over time.

Investor Cohort Analysis

Investors were grouped based on their first investment year.

Metrics calculated:

- Average SIP amount
- Total investment value
- Preferred funds

SIP Continuity Analysis

Investors with six or more SIP transactions were analyzed.

Investors having average gaps greater than 35 days were marked as:

At Risk Investors

Sector Concentration Analysis

Herfindahl-Hirschman Index (HHI) was calculated.

Formula:

HHI = $\Sigma(\text{weight}^2)$

Higher HHI indicates concentrated portfolio exposure.

Fund Recommendation System

A Python based recommender system was developed.

Input:

Low / Moderate / High Risk Appetite

Output:

Top 3 recommended funds based on Sharpe Ratio.

File:

recommender.py

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9. Power BI Dashboard

An interactive Power BI dashboard was created containing four analytical pages.

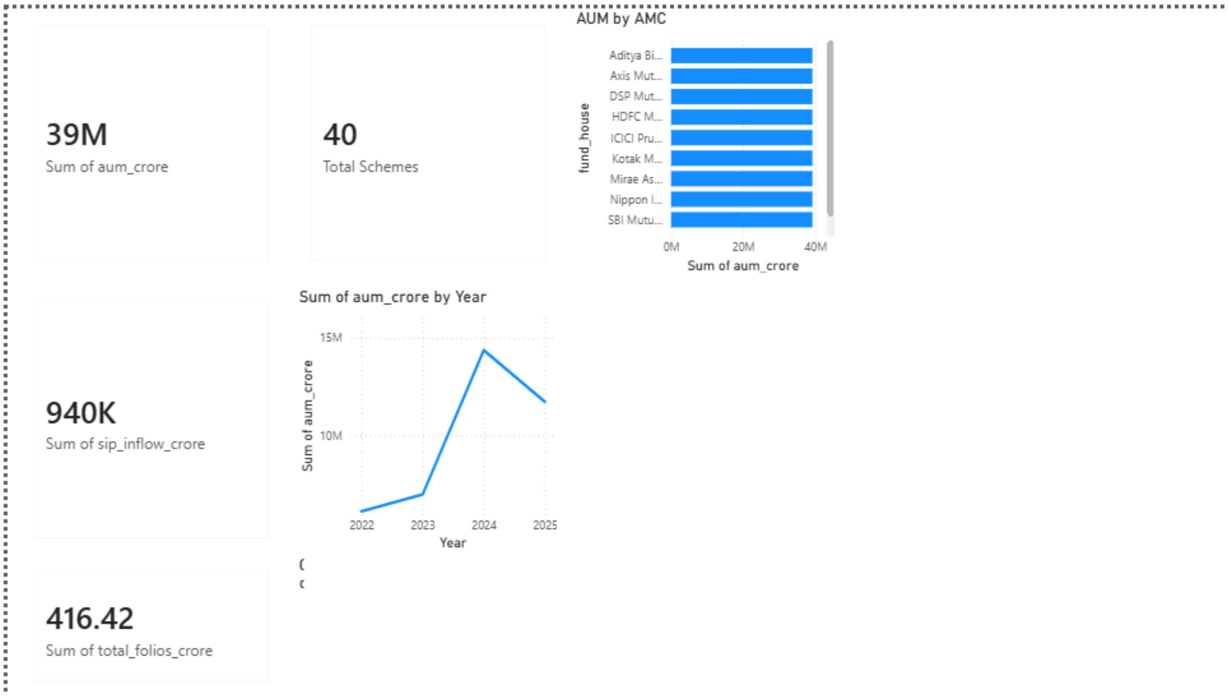
Dashboard File:

bluestock_mf_dashboard.pbix

Page 1: Industry Overview

Components:

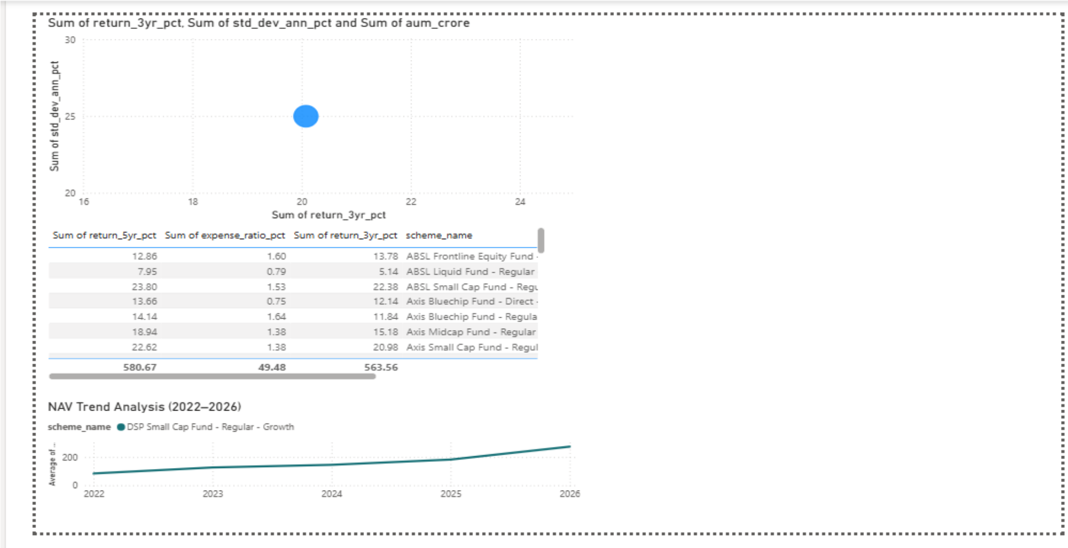
• Total AUM KPI
• SIP Inflow KPI
• Folio Count KPI
• Total Scheme KPI
• Industry AUM Trend
• AMC Comparison



Page 2: Fund Performance

Components:

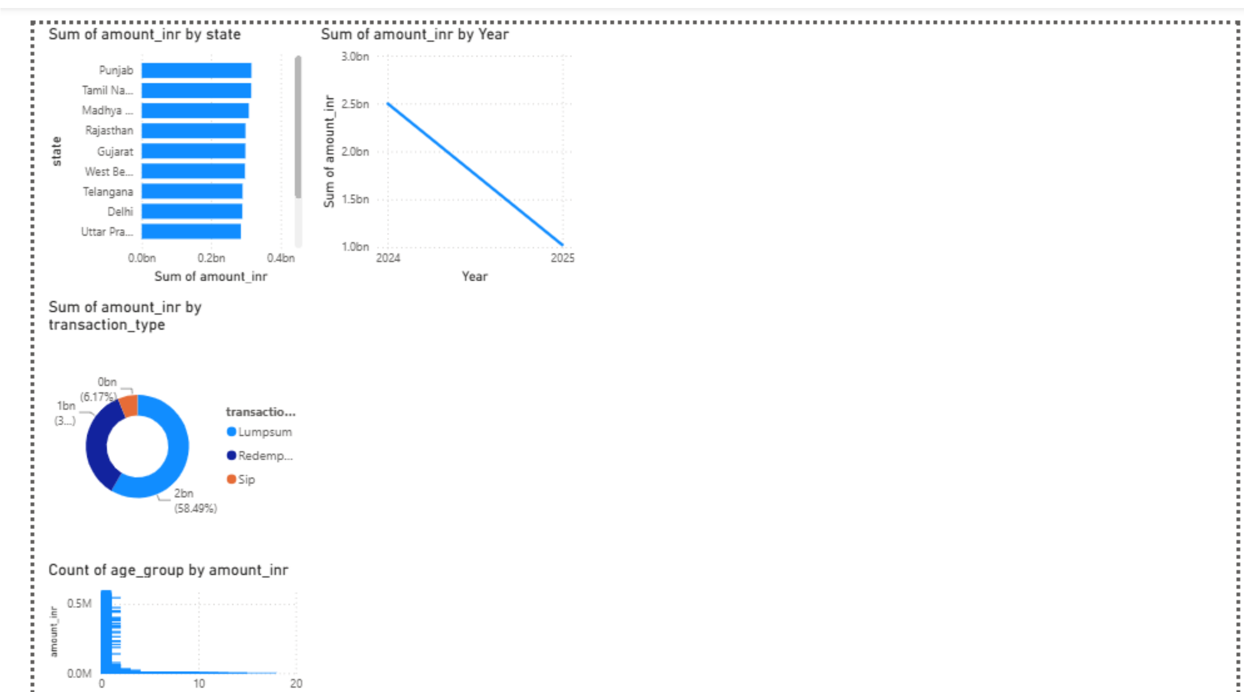
• Return vs Risk Scatter Plot
• Fund Scorecard Table
• NAV Trend
• Fund House Filter
• Category Filter
• Plan Filter



Page 3: Investor Analytics

Components:

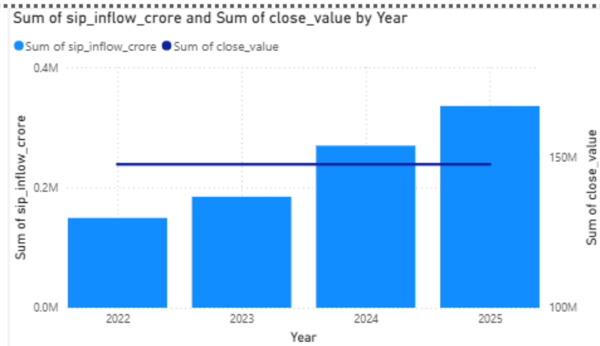
- State wise transaction analysis
- Transaction type split
- Age group investment analysis
- Monthly transaction trend



Page 4: SIP and Market Trends

Components:

• SIP inflow vs Market trend
• Category inflow heatmap
• Top performing categories



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10. Key Business Insights

1. SIP investment participation increased consistently over the analysis period.
2. Funds with higher Sharpe Ratio delivered better risk-adjusted performance.
3. Investor behaviour analysis helped identify SIP discontinuation risk.
4. Expense ratio plays an important role in long-term fund performance.
5. Diversified sector allocation reduces portfolio concentration risk.
6. Fund ranking models help simplify investment decision-making.

11. Conclusion

The Mutual Fund Analytics Platform successfully demonstrates a complete financial analytics solution by combining data engineering, database management, statistical analysis, risk modelling, and business intelligence.

The project transforms raw mutual fund datasets into actionable insights through automated pipelines, analytical models, and interactive dashboards.

The final solution enables investors and analysts to compare mutual funds, evaluate risks, identify trends, and make data-driven investment decisions.